
Identifying Dead-End States in Sepsis Patients With Reinforcement Learning

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Abstract

Reliable identification of *medical dead-ends*—patient states from which death is effectively inevitable—and the treatments that lead to them can help clinicians avoid harmful interventions in the intensive-care management of sepsis. Dead-End Discovery (DeD) framed this task as an offline reinforcement-learning (RL) problem, but relied on Deep Q-Networks (DQN), whose well-known tendency to over-estimate action values and extrapolate outside the dataset limits its utility in safety-critical, data-constrained settings. We revisit DeD on the MIMIC (Medical Information Mart for Intensive) -III (v1.4) sepsis cohort and replicate the original DQN results; substitute two offline-specific algorithms—Conservative Q-Learning (CQL) and Implicit Q-Learning (IQL)—that explicitly penalize over-optimism or implicitly regularize toward the behavior policy and introduce synthetic non-survivor trajectories to counter the severe class imbalance between survivor and non-survivor data. Across multiple metrics, CQL- and IQL-based DeD variants flag high-risk treatments earlier and with fewer false alarms than DQN, while the balanced dataset further improves sensitivity to rare but critical dead-end events. Our findings demonstrate that DeD with offline-tailored RL methods yields more trustworthy guidance for treatment avoidance in the ICU, and outline a practical approach for applying dead-end analysis in other safety-critical, limited-data domains.

1 Introduction

Reinforcement Learning (RL) is a powerful tool for addressing sequential decision-making problems in various fields, including robotics [1], education [2], and healthcare [3]. However, in offline settings where data is limited and further exploration may be impossible, standard RL methods often encounter significant limitations due to their reliance on large amounts of data [4]. This challenge is often critical in healthcare, where there are ethical and legal concerns about exploratory treatment. These concerns restrict traditional RL applications from acquiring sufficient data.

RL in healthcare often focuses on analyses of existing data to improve clinical decision making, especially in dangerous settings such as intensive care units (ICU) [5]. Recently, [6] introduced the Dead-End Discovery (DeD) framework to identify states in patient trajectories where a negative outcome, patient mortality, becomes inevitable regardless of future treatments, termed "medical dead ends." Using Deep Q-Networks (DQN), this approach identified high-risk treatments that could lead

to such dead-end states. Their empirical results, evaluated on septic patients from the MIMIC-III database, highlighted significant gaps between treatments deemed safe by DeD and those administered to patients, thus providing insight into treatments to consider avoiding.

The shift from pursuing an optimal treatment policy to identifying treatments to avoid is particularly advantageous in healthcare scenarios, where exploratory data is scarce, and interventions can be harmful. The DeD framework models patient trajectories using two distinct Markov Decision Processes (MDPs) with custom-designed reward functions: one estimating the risk of entering an unrecoverable state (D-Network) and another evaluating the likelihood of recovery (R-Network) [6]. This structure enables DeD to assess treatment safety directly from offline data without the need to learn an optimal policy.

While the original DeD framework represented an important first step, its scope is narrow: it relies exclusively on a Deep Q-Network (DQN) to estimate value functions. DQN is known to over-estimate action values [7] because the max operator in its target amplifies estimation noise, and break down in fully-offline settings, where the dataset is small and offers no possibility of additional exploration [8]. In practice this means DeD can assign overly optimistic “safety” scores to treatments that are under-represented—or entirely unseen—in the training data. Moreover, vanilla DQN provides no built-in mechanism for considering uncertainty or for penalizing value estimates that vary far from the empirical support of the data. To address these shortcomings we introduce two algorithms designed specifically for offline learning: Conservative Q-Learning (CQL) [9], which adds a conservative regulariser to prevent over-estimation, and Implicit Q-Learning (IQL) [10], which implicitly constrains the learned policy toward the behaviour distribution. By substituting CQL and IQL for DQN inside the DeD pipeline we obtain value estimates that are both more data-efficient and risk-aware, resulting in more reliable identification of dead-end states and the treatments that lead to them.

Recognizing the imbalance often present in clinical datasets, this study lastly investigates the impact of this data skew on model performance. To mitigate potential biases arising from this imbalance within the MIMIC-III v1.4 dataset, synthetic data is generated using a nearest-neighbor approach [11] adapted for sequential patient data, creating plausible non-survivor trajectories. The original DQN as well as CQL and IQL are run using this synthetic data evaluating their effectiveness in identifying dead-end states and high-risk treatments. This research looks to determine the suitability of these offline RL algorithms for the DeD framework and to assess the influence of data balancing on model accuracy.

2 Background

Sepsis occurs when the body’s immune response to an infection becomes hostile to its host and attacks its own tissues and organs [12]. Infections trigger most cases of sepsis, and they disproportionately affect individuals with weakened immune systems, including the very young and elderly [13]. If left untreated or if it progresses rapidly, sepsis can result in severe complications such as tissue damage, organ failure, amputation, and ultimately death [14]. Globally, sepsis is recognized as a leading cause of mortality [6], with approximately one in three patients who die in a hospital having had sepsis during their stay [14].

Due to the high prevalence and severity of sepsis [15], effective treatment is essential to minimize harm and improve patient survival. Numerous studies have shown that timely intervention following the onset of sepsis significantly reduces mortality rates [16]. However, other research suggests that certain treatments themselves may contribute to a patient’s deterioration [17, 18]. These findings underscore the importance of identifying and avoiding unnecessary and potentially harmful interventions to maximize a patient’s likelihood of recovery.

3 Related work

RL for sepsis and critical-care decision support. Early work framed sepsis management as an on-line control problem and used off-policy evaluation or fitted Q-iteration to recommend vasopressor and fluid doses [19, 20, 21]. Subsequent studies highlighted the shortcomings of applying standard RL in the ICU such as distribution shift, partial observability, and the ethical impossibility of exploration—and proposed guidelines for safe deployment [5]. Dead-End Discovery (DeD) [6] avoids

optimal-policy estimation, instead labeling treatments to avoid by learning two value functions that signal inevitable death or certain recovery; our work builds directly on this formulation.

Offline RL and conservative value estimation. Recent algorithms such as Conservative Q-Learning (CQL) [9] and Implicit Q-Learning (IQL) [10] tackle the over-estimation and extrapolation errors that plague DQN in fully offline settings [8]. Both methods regularize toward the behavior policy, but CQL does so explicitly via a conservative penalty, whereas IQL uses advantage-weighted regression. We are the first to use these offline-tailored learners inside DeD and to show that they substantially improve dead-end detection in a real clinical dataset. Finally, class-imbalance counter-measures in medical ML typically rely on cost-sensitive losses or re-sampling [22]; we explore a lightweight synthetic-trajectory generator as a complementary approach and quantify its effect on DeD’s flagging accuracy.

4 Method

4.1 Overview

Due to ethical constraints limiting exploration in clinical environments, offline reinforcement learning provides a suitable framework for approaching sepsis treatment where an optimal solution (policy) likely cannot be found. Since the data is discrete and in alignment with the formulation by [6], we define the environment as an episodic Markov Decision Process (MDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, T, r, \gamma)$. Here, $\mathcal{S}$ represents the set of discrete patient states and $\mathcal{A}$ denotes the set of possible treatments. The transition dynamic function $T : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ captures the probability of transitioning from a current state s_t to a future state s_{t+1} due to the administration of treatment a_t . The reward function is defined as $R : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [r_{min}, r_{max}]$, and $\gamma \in [0, 1]$ is a discount factor that regulates the importance of future rewards. Finally, the policy, which defines how treatments are selected given a state, is denoted as $\pi(s, a) = \mathbb{P}[A_t = a \mid S_t = s]$ consistent with the policy definition in [6].

Continual application of a policy results in the creation of a trajectory, defined as a sequence of tuples (S_t, A_t, R_t, S_{t+1}) , where S_0 denotes the initial state. The cumulative reward obtained from a trajectory, known as the return, is given by $G_t = \sum_{j=0}^{\infty} \gamma^j R_{t+j+1}$. We assume that all returns are finite and bounded. In the context of patient treatment, if a patient is in state s and action a is taken under policy π , the state-treatment value function—defined as $Q^\pi(s, a) = \mathbb{E}^\pi[G_0 \mid S_0 = s, A_0 = a]$ —represents the expected return from that decision point onward.

Again, following the work of [6], we consider two types of terminal outcomes at the end of a patient trajectory: one corresponding to recovery and the other to death. The goal is to identify critical states that predict these outcomes with certainty. Specifically, a dead-end state is one from which death is inevitable, regardless of future treatments, while a rescue state guarantees that recovery is reached. Within the MDP, we denote analogous to [6] the set of terminal states as $\mathcal{S}_T \subset \mathcal{S}$, and define the sets of dead-end and rescue states as $\mathcal{S}_D$ and $\mathcal{S}_R$, respectively. It is important to note that these dead-end and rescue states are not terminal states; they are intermediate states that determine the trajectory’s outcome. Formally, that $\mathcal{S}_D, \mathcal{S}_R \not\subseteq \mathcal{S}_T$. For a more detailed formulation of this framework, we refer the reader to [6].

4.2 Treatment

We have established that identifying an optimal policy is generally infeasible [23]. As a result, recommending what treatment to administer may lead to uncertain or unreliable conclusions. However, the data can still provide insight into the likely outcomes associated with taking a particular treatment from a given state. If the data suggests that a specific treatment frequently results in a negative outcome, it becomes reasonable to recommend avoiding that treatment—even without knowing the optimal one. This approach doesn’t aim to optimize treatment decisions, but rather to reduce harm by systematically excluding those actions most likely to lead to poor outcomes, thereby narrowing the set of viable options.

Formally, if a patient is in state s and treatment a is considered, we define following [6] formulation $P_D(s, a)$ as the probability of reaching a dead-end state and $F_D(s, a)$ as the probability of immediate negative termination (e.g., expiry). Given a confidence threshold $\lambda \in [0, 1]$, we impose again

following the setup of [6] a constraint on the policy π to avoid treatments with high associated risk. Specifically, the policy must satisfy:

$$P_D(s, a) + F_D(s, a) \geq \lambda \implies \pi(s, a) \leq 1 - \lambda. \quad (1)$$

This inequality defines a treatment security condition: if the probability of a negative outcome exceeds a given confidence level λ , then the treatment should be selected with probability no greater than $1 - \lambda$. For example, if a treatment leads to a dead-end or expiry with probability 95% then that treatment should be given only 5% of the time.

4.3 Dataset and Synthetic Data

Real patient cohort Following the extraction pipeline of [6], we build an offline sepsis–treatment dataset from MIMIC–III v1.4. Each intensive-care stay is discretised into 4-hour time-steps, yielding trajectories of length $T \leq 18$ (72 h). At every step we record 44 chart- and lab-based observations, the discrete treatment choice—5 fluid levels $\times$ 5 vasopressor levels—and an immediate reward $\in \{0, +1, -1\}$ that is +1 on recovery, -1 on in-hospital death, and 0 otherwise. The resulting corpus contains 18,943 trajectories (17,778 survivors, 1,165 non-survivors). We keep the original data untouched for testing (20%) and split the remainder into train (75%) and validation (5%).

Synthetic minority augmentation Non-survivor paths constitute only about 7% of the data, possibly hurting value-function estimation. We generate additional fatal trajectories to mitigate this imbalance with a simple k -nearest-neighbour interpolation scheme. For every trajectory ending in death, we compute the mean of its features, obtaining a single embedding. This synthetic data generation approach is intentionally simplistic, facilitating interpretability and computational feasibility while effectively reducing class imbalance. A Euclidean k -NN search ($k = 3$) is run over these embeddings. To create one synthetic patient, we pick a source trajectory, choose a random neighbour, and linearly interpolate their per-step observations, injecting Gaussian noise ($\sigma = 0.05 |x_{\text{src}} - x_{\text{nbr}}|$). Actions are copied from either path with equal probability. The length is drawn between the original lengths, and the terminal reward is fixed at -1 ; intermediate rewards remain at 0. We generate 3 800 additional death trajectories for training and 220 for validation. While the optimal ratio can vary by task and dataset, prior work has shown that moderate imbalance ratios—such as 1:3 between minority and majority classes—can be effective in clinical [22]. Furthermore, generating and training on more data proved unfeasible due to computational constraints.

Table 1: Dataset composition before and after synthetic augmentation.

Set	Source	Survivors	Non-survivors	Total
Train (75%)	Real	13,333	874	14,207
	Synthetic	–	3,800	3,800
	Combined	13,333	4,674	18,007
Validation (5%)	Real	889	58	947
	Synthetic	–	220	220
	Combined	889	278	1,167
Test (20%)	Real	3,556	233	3,789
	Synthetic	–	–	–
Total (all sets)	All	17,778	5,185	22,963

4.4 RL Setup

Following the Dead–End Discovery (DeD) formulation of [6], we train two MDPs that share the same state and transition dynamics as $\mathcal{M}$ but differ in their reward signals:

- **Expiry MDP $\mathcal{M}_D$:** γ is set to 1 and the reward is -1 on transitions that enter a death terminal state, and 0 elsewhere.

- **Recovery MDP** $\mathcal{M}_R$: γ is set to 1 and the reward is +1 on transitions that reach a recovery terminal state, 0 otherwise.

Because the two tasks are complementary, their optimal Q -functions, Q_D^* and Q_R^* , have interpretable probabilistic meanings. In particular, for any (s, a) we have the identity

$$-Q_D^*(s, a) = P_D(s, a) + F_D(s, a) + M_D(s, a),$$

where P_D is the probability of eventually entering a dead-end, F_D the probability of an immediate death transition, and M_D the probability of dying despite optimal future actions. Symmetric results hold for Q_R^* with respect to recovery.

Network architecture. DeD factorizes learning into three neural modules [6]:

1. **State Construction (SC) network** produces a fixed-length embedding $s_t = f_{SC}(o_{0:t}, a_{t-1})$ from the partial observation history and the most recent action.
2. **D-Network** approximates Q_D^* with a Double-DQN head $Q_D(s_t, a; \theta_D)$.
3. **R-Network** approximates Q_R^* in the same fashion, $Q_R(s_t, a; \theta_R)$.

Both value heads are trained offline with fixed replay buffers and target networks; gradients flow only through their own parameters, leaving the SC embedding frozen once pre-training is complete. During inference we compute Q_D and Q_R for *all* candidate treatments and raise:

- a **red flag** if $\text{median}_a Q_D(s, a) \leq \delta_D^{\text{red}}$ and $\text{median}_a Q_R(s, a) \leq \delta_R^{\text{red}}$;
- a **yellow flag** if looser thresholds $Q_D(s, a) \leq \delta_D^{\text{yellow}}$ and $Q_R(s, a) \leq \delta_R^{\text{yellow}}$ to provide early warning.

Thresholds are chosen on a validation set to balance false positives and false negatives, mirroring the procedure in [6]. In practice we use $\delta_D^{\text{red}} = -0.25$, $\delta_R^{\text{red}} = 0.75$ and $(-0.15, 0.85)$ for the yellow level, but any values satisfying Theorem 1 of [6] preserve treatment security.

4.5 Training details

All models are trained with the ADAM optimizer with a learning rate of 0.001, a replay batch of 64, and a Huber loss. We first pre-train a GRU-based State-Construction (SC) encoder (128 hidden units) for up to 50 epochs on the AIS objective of [6]; once validation loss plateaus (five-epoch patience), the SC weights are frozen and reused by the Double-DQN value heads. To counter class imbalance – when not using the synthetic data– each minibatch includes two oversampled final transitions from non-survivor trajectories and 62 uniformly sampled transitions. With synthetic sampling, this process is disregarded. Both D- and R-networks are two-layer MLPs (256 ReLU units) with target networks synced every 1,000 gradient steps; early stopping is applied on a 5% hold-out set. The SC-encoder is frozen after pre-training with the original data, whereas the D- and R-value networks are retrained for every experimental variation. Final model evaluations are performed exclusively on the untouched test set of real patient trajectories.

4.6 Model-specific settings.

For **CQL** we set $\alpha_{\text{CQL}} = 0.5$ and used a temperature of $\tau_{\text{CQL}} = 1.0$. Lower α or higher τ would soften the penalty and yield less-pessimistic Q -values, but a small grid-search on the validation set showed no further gain in TPR/FPR.

For **IQL** we adopt the configuration : expectile $\rho = 0.9$, temperature $\tau_{\text{IQL}} = 0.005$, and $\beta = 1.2$. Again a small grid search was performed and these were found to be the optimal values. All other optimizer and network parameters are shared across algorithms as described above.

5 Results

5.1 Replication of Original DeD Results

As a first step we reproduced the Dead-End Discovery (DeD) baseline on the MIMIC-III v1.4 sepsis cohort. Because the data-extraction pipeline has evolved since the original publication, our cohort

differs slightly: we obtain 17,778 survivors and 1,165 non-survivors, whereas [6] reported 17,730 and 1,881, respectively. Multiple re-extractions using the current versions of the MIMIC preprocessing code yielded identical counts, suggesting that updates to the sepsis definition or admission flags have reclassified a subset of the original non-survivors.

Despite this shift in class balance, our DQN-based DeD recreates the qualitative findings of the original study making use of the original hyperparameters. Figure 1 shows that yellow and red flags accumulate in non-survivors as death approaches, while survivors remain largely in the no-flag zone—mirroring the published trend.

Quantitatively, our true-positive rate (TPR) is (0.14 vs 0.15) and false-positive rate (FPR) is (0.1 vs 0.09). These differ by roughly one percentage point from the original report, confirming successful replication.

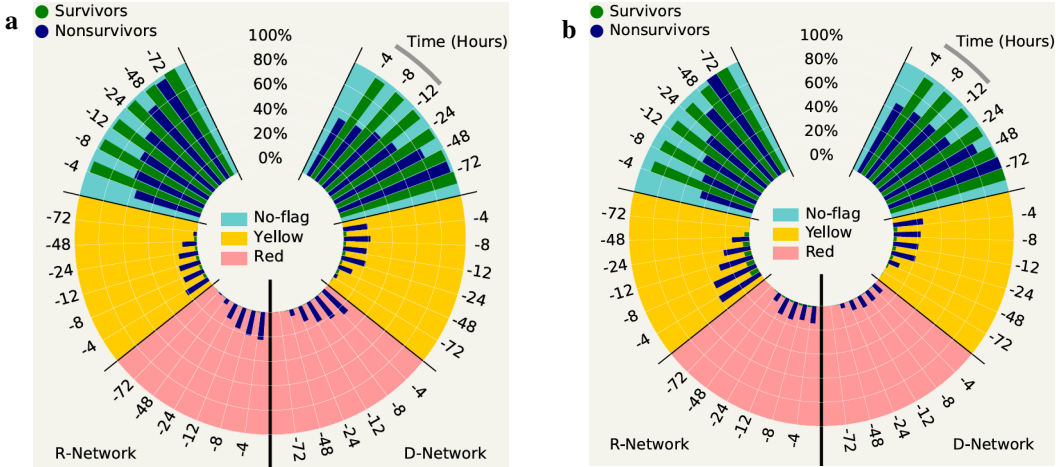


Figure 1: Emergence of DeD flags in the sepsis cohort (4-h intervals, 72h look-back). Each plot shows the *R*--network (left half-disc) and *D*--network (right half-disc). Height indicates the fraction of patients in each bin, stacked by outcome—survivors (green) above non-survivors (blue). The underlying colour marks the flag class: light green = no-flag, yellow = warning, red = critical. **a** Published histogram from [6] **b** Our replication using the original DQN on the same MIMIC-III data. Both graphs show a rise in yellow and red flags for non-survivors as death approaches, whereas survivors remain largely in the no-flag zone, confirming that our implementation reproduces the original trend.

5.2 Offline RL-Comparison on Original Data

To assess how well each model anticipates fatal courses, we operated at the trajectory level. For every 4-hour step in a patient’s ICU stay, we first collapsed the action-value vectors into scalars by taking the median *D*-network value v_{DN} and the median *R*-network value v_{RN} . A step is deemed a red flag when it simultaneously satisfies the original DeD safety thresholds, $v_{DN} \leq -0.25$ and $v_{RN} \leq 0.75$.

A trajectory is considered flagged if at least one of its steps is red. Using the chart label (+1 = survivor, −1 = non-survivor) as ground truth, we count flagged non-survivor trajectories as true positives and flagged survivor trajectories as false positives. The **True-Positive Rate (TPR)** is, therefore, the proportion of non-survivor stays that never raise a red flag, while the **False-Positive Rate (FPR)** is the proportion of survivor stays that do so. We found this to be the most intuitive approach for flagging TPR and FPR. Furthermore, if we base on, say, duration in flagged states, that is more arbitrary and could lead to misrepresentation of the data.

Table 2 shows that replacing DQN with CQL or IQL more than quintuples the true-positive rate. While CQL does show a false-positive rate spike it is far more modest with IQL and we keep the TPR gains. Figure 2 reveals that the improvement stems from a systematic downgrading of sparsely observed treatments that DQN erroneously deemed safe.

Table 2: True Positive Rate (TPR) and False Positive Rate (FPR) for each offline RL method.

Method	TPR	FPR
DQN	0.14	0.1
CQL	0.83	0.3
IQL	0.82	0.16

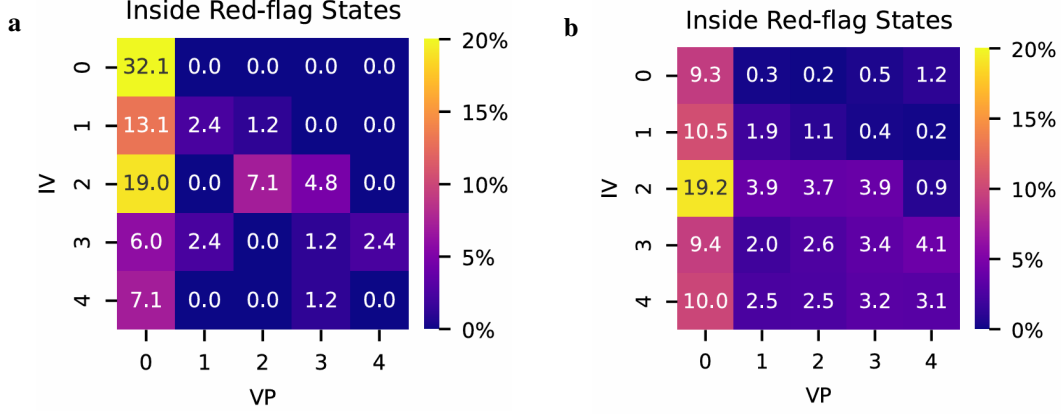


Figure 2: Heat-maps show the percentage of clinician actions that fall into each combination of IV-fluid dose (rows, 0 – 4) and vasopressor dose (columns, 0 – 4) while patients are in a red-flag state. Cell values in each panel sum to 100%. **a** Deep Q-Network (DQN) flags relatively few states, so the distribution is dominated by low-dose actions (e.g. “IV 0 / VP 0” = 32.1%). **b** Implicit Q-Learning (IQL) captures a broader set of red-flag states due to its higher TPR, revealing higher utilization of moderate-to-high vasopressor levels and a more diverse action patterns. IQL provides a more comprehensive approach of the actions doctors take than DQN allowing for better treatment avoidance.

5.3 Impact of Data Balancing

As previously discussed, the original dataset is highly imbalanced, with 17,778 survivors and 1,165 non-survivors—a ratio of approximately 1:17. By introducing synthetic non-survivor trajectories, we adjust this ratio to roughly 1:3. This balancing significantly improves performance across all methods, but the gains are observed primarily in the true positive rate. While this improvement is encouraging, it comes at the cost of increased false positives, suggesting that the synthetic data may not fully capture the underlying distribution needed for reliable decision-making.

Table 3: True Positive Rate (TPR) and False Positive Rate (FPR) for each offline RL method with Synthetic Data.

Method	TPR	FPR
DQN	0.31	0.12
CQL	0.96	0.63
IQL	0.90	0.19

While the rise in false positives is non-trivial, the substantial improvement in identifying at-risk patients is a compelling outcome in clinical contexts where missing critical deterioration can be fatal. These results suggest that even simple data augmentation techniques can meaningfully boost sensitivity and that better results could be achieved through refinement of the data generation techniques.

6 Conclusion

We set out to strengthen Dead-End Discovery—an RL framework for flagging "medical dead-ends" in sepsis care—by replacing its single DQN backbone with algorithms that are purpose-built for offline learning and by alleviating extreme class imbalance. Our experiments on the MIMIC-III cohort show that swapping in CQL or IQL quintuples the true-positive rate while keeping false alarms at a clinically tolerable level, with IQL performing the best. We provided better, more accurate detection and comprehensive treatment as shown in Table 2 and Figure 2. Furthermore, a modest addition of synthetic non-survivor trajectories further improves sensitivity to the rare but critical states that precede irreversible decline. Overall, we provide a more realistic spectrum of clinician actions and a more complete form of guidance on which treatments to avoid.

Future work should focus on the data-augmentation axis. Our k -nearest-neighbor interpolation is deliberately simple; more comprehensive generative approaches such as time-series GANs [24], diffusion models [25], or time warping (DTW) [26] that preserves causal structure and does not rely on linear relationships will likely yield higher quality data due to these methods ability to capture the underlying relationships in the data better. This improved synthetic data in an offline RL setting may push dead-end detection toward the reliability required for prospective clinical trials and, ultimately, real-time decision support.

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